

Subnetting Practice

Host Requirements · CIDR Notation · IPv4

1

200.1.1.0/24 → 20 Hosts per Network

Class C · Original prefix /24

1 FIND THE REQUIRED HOST BITS ($2^h - 2 \geq \text{HOSTS NEEDED}$)

Need ≥ 20 usable hosts per network

Formula: $2^h - 2 \geq 20$

Try $h = 4$: $2^4 - 2 = 16 - 2 = 14$ ✗ not enough

Try $h = 5$: $2^5 - 2 = 32 - 2 = 30$ ✓ sufficient

∴ Host bits $h = 5$

2 CALCULATE THE NEW PREFIX LENGTH ($32 - h$)

Original prefix = /24 (24 network bits)

New prefix = $32 - h = 32 - 5 = 27$

∴ New CIDR notation = /27

3 DERIVE THE SUBNET MASK (BINARY → DECIMAL)

/27 binary: 11111111 . 11111111 . 11111111 . 11100000

 255 . 255 . 255 . ?

4th octet: $128 + 64 + 32 = 224$

∴ Subnet mask = 255 . 255 . 255 . 224

4 BLOCK SIZE, SUBNET COUNT, AND USABLE HOSTS

Block size = $2^h = 2^5 = 32$

Borrowed bits = new prefix - orig = $27 - 24 = 3$

Number of subnets = $2^3 = 8$

Usable hosts/net = $2^5 - 2 = 30$

STEP 5 · ALL 8 SUBNETS (BLOCK SIZE = 32)

#	NETWORK ADDRESS	FIRST HOST	LAST HOST	BROADCAST
1	200.1.1.0/27	200.1.1.1	200.1.1.30	200.1.1.31

#	NETWORK ADDRESS	FIRST HOST	LAST HOST	BROADCAST
2	200.1.1.32/27	200.1.1.33	200.1.1.62	200.1.1.63
3	200.1.1.64/27	200.1.1.65	200.1.1.94	200.1.1.95
4	200.1.1.96/27	200.1.1.97	200.1.1.126	200.1.1.127
5	200.1.1.128/27	200.1.1.129	200.1.1.158	200.1.1.159
6	200.1.1.160/27	200.1.1.161	200.1.1.190	200.1.1.191
7	200.1.1.192/27	200.1.1.193	200.1.1.222	200.1.1.223
8	200.1.1.224/27	200.1.1.225	200.1.1.254	200.1.1.255

2

199.9.10.0/24 → 12 Hosts per Network

Class C · Original prefix /24

1 FIND THE REQUIRED HOST BITS ($2^h - 2 \geq \text{HOSTS NEEDED}$)

Need ≥ 12 usable hosts per network

Formula: $2^h - 2 \geq 12$

Try $h = 3$: $2^3 - 2 = 8 - 2 = 6$ ✗ not enough

Try $h = 4$: $2^4 - 2 = 16 - 2 = 14$ ✓ sufficient

∴ Host bits $h = 4$

2 CALCULATE THE NEW PREFIX LENGTH ($32 - h$)

Original prefix = /24 (24 network bits)

New prefix = $32 - h = 32 - 4 = 28$

∴ New CIDR notation = /28

3 DERIVE THE SUBNET MASK (BINARY → DECIMAL)

/28 binary: 11111111 . 11111111 . 11111111 . 11110000

255 255 255 ?

4th octet: $128 + 64 + 32 + 16 = 240$

∴ Subnet mask = 255 . 255 . 255 . 240

4 BLOCK SIZE, SUBNET COUNT, AND USABLE HOSTS

Block size = $2^h = 2^4 = 16$

Borrowed bits = new prefix - orig = $28 - 24 = 4$

Number of subnets = $2^4 = 16$

Usable hosts/net = $2^4 - 2 = 14$

STEP 5 · ALL 16 SUBNETS (BLOCK SIZE = 16)

#	NETWORK ADDRESS	FIRST HOST	LAST HOST	BROADCAST
1	199.9.10.0/28	199.9.10.1	199.9.10.14	199.9.10.15
2	199.9.10.16/28	199.9.10.17	199.9.10.30	199.9.10.31
3	199.9.10.32/28	199.9.10.33	199.9.10.46	199.9.10.47
4	199.9.10.48/28	199.9.10.49	199.9.10.62	199.9.10.63
5	199.9.10.64/28	199.9.10.65	199.9.10.78	199.9.10.79
6	199.9.10.80/28	199.9.10.81	199.9.10.94	199.9.10.95
7	199.9.10.96/28	199.9.10.97	199.9.10.110	199.9.10.111
8	199.9.10.112/28	199.9.10.113	199.9.10.126	199.9.10.127
9	199.9.10.128/28	199.9.10.129	199.9.10.142	199.9.10.143
10	199.9.10.144/28	199.9.10.145	199.9.10.158	199.9.10.159
11	199.9.10.160/28	199.9.10.161	199.9.10.174	199.9.10.175
12	199.9.10.176/28	199.9.10.177	199.9.10.190	199.9.10.191
13	199.9.10.192/28	199.9.10.193	199.9.10.206	199.9.10.207
14	199.9.10.208/28	199.9.10.209	199.9.10.222	199.9.10.223
15	199.9.10.224/28	199.9.10.225	199.9.10.238	199.9.10.239
16	199.9.10.240/28	199.9.10.241	199.9.10.254	199.9.10.255

3

172.50.0.0/16 → 1000 Hosts per Network

Class B · Original prefix /16

1 FIND THE REQUIRED HOST BITS ($2^h - 2 \geq \text{HOSTS NEEDED}$)

Need ≥ 1000 usable hosts per network

Formula: $2^h - 2 \geq 1000$

Try $h = 9$: $2^9 - 2 = 512 - 2 = 510$ ✗ not enough

Try $h = 10$: $2^{10} - 2 = 1024 - 2 = 1022$ ✓ sufficient

∴ Host bits $h = 10$

2 CALCULATE THE NEW PREFIX LENGTH ($32 - h$)

Original prefix = /16 (16 network bits)

New prefix = $32 - h = 32 - 10 = 22$

∴ New CIDR notation = /22

3 DERIVE THE SUBNET MASK (BINARY → DECIMAL)

/22 binary: 11111111 . 11111111 . 11111100 . 00000000

255 255 ? 0

3rd octet: $128+64+32+16+8+4 = 252$

$\therefore$ Subnet mask = 255 . 255 . 252 . 0

4 BLOCK SIZE, SUBNET COUNT, AND USABLE HOSTS

Block size = $2^h = 2^{10} = 1024$ (spans +4 in 3rd octet)

Borrowed bits = $22 - 16 = 6$

Number of subnets = $2^6 = 64$

Usable hosts/net = $2^{10} - 2 = 1022$

STEP 5 · FIRST 5 AND LAST OF 64 SUBNETS (BLOCK = +4 IN 3RD OCTET)

#	NETWORK ADDRESS	FIRST HOST	LAST HOST	BROADCAST
1	172.50.0.0/22	172.50.0.1	172.50.3.254	172.50.3.255
2	172.50.4.0/22	172.50.4.1	172.50.7.254	172.50.7.255
3	172.50.8.0/22	172.50.8.1	172.50.11.254	172.50.11.255
4	172.50.12.0/22	172.50.12.1	172.50.15.254	172.50.15.255
5	172.50.16.0/22	172.50.16.1	172.50.19.254	172.50.19.255
...	... /22	...	...	...
64	172.50.252.0/22	172.50.252.1	172.50.255.254	172.50.255.255

Full pattern: 172.50.X.0/22 where X = 0, 4, 8, 12, ..., 252

4

12.0.0.0/8 → 100 Hosts per Network

Class A · Original prefix /8

1 FIND THE REQUIRED HOST BITS ($2^h - 2 \geq$ HOSTS NEEDED)

Need ≥ 100 usable hosts per network

Formula: $2^h - 2 \geq 100$

Try h = 6: $2^6 - 2 = 64 - 2 = 62$ x not enough

Try h = 7: $2^7 - 2 = 128 - 2 = 126$ ✓ sufficient

$\therefore$ Host bits h = 7

2 CALCULATE THE NEW PREFIX LENGTH (32 - H)

Original prefix = /8 (8 network bits)

New prefix = $32 - h = 32 - 7 = 25$

$\therefore$ New CIDR notation = /25

3 DERIVE THE SUBNET MASK (BINARY → DECIMAL)

/25 binary: 11111111 . 11111111 . 11111111 . 10000000

4th octet: 128 = 128

∴ Subnet mask = 255 . 255 . 255 . 128

4 BLOCK SIZE, SUBNET COUNT, AND USABLE HOSTS

Block size = 2^h = 2^7 = 128

Borrowed bits = new prefix - orig = 25 - 8 = 17

Number of subnets = 2^{17} = 131,072

Usable hosts/net = $2^7 - 2$ = 126

STEP 5 · FIRST 4 AND LAST OF 131,072 SUBNETS (BLOCK = +128 IN 4TH OCTET)

#	NETWORK ADDRESS	FIRST HOST	LAST HOST	BROADCAST
1	12.0.0.0/25	12.0.0.1	12.0.0.126	12.0.0.127
2	12.0.0.128/25	12.0.0.129	12.0.0.254	12.0.0.255
3	12.0.1.0/25	12.0.1.1	12.0.1.126	12.0.1.127
4	12.0.1.128/25	12.0.1.129	12.0.1.254	12.0.1.255
...	... /25	...	...	...
131,072	12.255.255.128/25	12.255.255.129	12.255.255.254	12.255.255.255

Full pattern: 12.A.B.0/25 and 12.A.B.128/25 for every A,B in 0-255

■ FINAL ANSWERS · ALL FOUR PROBLEMS SOLVED ■

#	NETWORK	NEED	H BITS	PREFIX	SUBNET MASK	BLOCK	SUBNETS	USABLE
1	200.1.1.0/24	20	5	/27	255.255.255.224	32	8	30
2	199.9.10.0/24	12	4	/28	255.255.255.240	16	16	14
3	172.50.0.0/16	1000	10	/22	255.255.252.0	1024	64	1022
4	12.0.0.0/8	100	7	/25	255.255.255.128	128	131,072	126

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